

# PADHAISHURU

GATE Preparation Platform

## GATE TF

Textile Engineering & Fibre Science

### TREND-BASED MOCK PAPER

— 01 —

|                 |             |
|-----------------|-------------|
| Duration        | 3 Hours     |
| Total Questions | 65          |
| Maximum Marks   | 100         |
| Sections        | 10 subjects |

#### Difficulty Distribution

Easy: 14 questions (22%)

Moderate: 42 questions (65%)

Difficult: 9 questions (14%)

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.  
These mock papers are constructed using analysis of historical GATE  
PYQ patterns.  
Historical patterns do not guarantee future examination questions.

# GENERAL INSTRUCTIONS

GATE TF — Trend-Based Mock Paper

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## 1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

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## 2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
  - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
  - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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## 3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
  - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
  - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
  - NAT questions carry no negative marking.
  - MSQ questions carry no negative marking.
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## 4. How to Answer

- For MCQ: Select exactly one option.
  - For MSQ: Select one or more options. Partial credit is NOT awarded.
  - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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## 5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

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## 6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

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## 7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
  - The Question Booklet contains multiple pages. Verify the page count.
  - Immediately report any discrepancy in the Question Booklet to the invigilator.
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**Q1.**

1M [no -ve]

*[General Aptitude]*

Perimeter of square side 5:

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*Enter your answer here***Q2.**

1M [-0.33]

*[Engineering Mathematics]*Integral of  $2x \, dx$ :

- (A)  $x^2 + C$
- (B)  $2x + C$
- (C)  $x + C$
- (D)  $x^2/2$

**Q3.**

1M [-0.33]

*[Textile Fibers]*

Cotton is a:

- (A) Protein
- (B) Cellulosic
- (C) Synthetic
- (D) Mineral

**Q4.**

1M [-0.33]

*[Yarn Manufacturing]*

Carding does:

- (A) Parallelize fibers
- (B) Spin fibers
- (C) Dye
- (D) Weave

**Q5.**

1M [-0.33]

*[Fabric Manufacturing]*

Plain weave has:

- (A) 1 up 1 down
- (B) 2 up 2 down
- (C) 3 up 1 down
- (D) Random

**Q6.**

1M [-0.33]

*[Textile Testing]*

Tensile strength measured by:

- (A) Microscope
- (B) Tensile tester
- (C) Air permeability
- (D) None

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Q7. 1M [-0.33]*[Chemical Processing]*

Reactive dye bonds with:

- (A) Physical
- (B) Covalent
- (C) Hydrogen
- (D) Van der Waals

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Q8. 1M [-0.33]*[Apparel Engineering]*

Pattern is for:

- (A) Fabric design
- (B) Garment shape
- (C) Dyeing
- (D) Finishing

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Q9. 1M [-0.33]*[Textile Machinery]*

Ring frame produces:

- (A) Sliver
- (B) Roving
- (C) Yarn
- (D) Fabric

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Q10. 1M [-0.33]*[Textile Physics]*Coulomb friction law:  $F = ?$ 

- (A)  $\mu N$
- (B)  $\mu^2 N$
- (C)  $N/\mu$
- (D)  $\mu^2 N$

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Q11. 1M [no -ve]*[Yarn Manufacturing]*

Carding does:

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*Enter your answer here*

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Q12. 1M [no -ve]*[Textile Machinery]*

Ring frame produces:

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*Enter your answer here*

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Q13. 1M [-0.33]*[General Aptitude]*

Perimeter of square side 5:

- (A) 10
- (B) 20
- (C) 25
- (D) 5

**Q14.**

1M [-0.33]

*[Chemical Processing]*

Reactive dye bonds with:

- (A) Physical
- (B) Covalent
- (C) Hydrogen
- (D) Van der Waals

**Q15.**

1M [-0.33]

*[Engineering Mathematics]*Integral of  $2x \, dx$ :

- (A)  $x^2 + C$
- (B)  $2x + C$
- (C)  $x + C$
- (D)  $x^2/2$

**Q16.**

1M [-0.33]

*[Engineering Mathematics]*Integral of  $2x \, dx$ :

- (A)  $x^2 + C$
- (B)  $2x + C$
- (C)  $x + C$
- (D)  $x^2/2$

**Q17.**

1M [no -ve]

*[Textile Machinery]*

Ring frame produces:

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*Enter your answer here***Q18.**

1M [-0.33]

*[General Aptitude]*

Perimeter of square side 5:

- (A) 10
- (B) 20
- (C) 25
- (D) 5

**Q19.**

1M [-0.33]

*[Engineering Mathematics]*Integral of  $2x \, dx$ :

- (A)  $x^2 + C$
- (B)  $2x + C$
- (C)  $x + C$
- (D)  $x^2/2$

**Q20.**

1M [-0.33]

*[Chemical Processing]*

Reactive dye bonds with:

- (A) Physical
- (B) Covalent
- (C) Hydrogen
- (D) Van der Waals

**Q21.**

1M [-0.33]

*[Fabric Manufacturing]*

Plain weave has:

- (A) 1 up 1 down
- (B) 2 up 2 down
- (C) 3 up 1 down
- (D) Random

**Q22.**

1M [no -ve]

*[Textile Machinery]*

Ring frame produces:

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*Enter your answer here***Q23.**

1M [-0.33]

*[Textile Machinery]*

Ring frame produces:

- (A) Sliver
- (B) Roving
- (C) Yarn
- (D) Fabric

**Q24.**

1M [-0.33]

*[Textile Machinery]*

Ring frame produces:

- (A) Sliver
- (B) Roving
- (C) Yarn
- (D) Fabric

**Q25.**

1M [-0.33]

*[Chemical Processing]*

Reactive dye bonds with:

- (A) Physical
- (B) Covalent
- (C) Hydrogen
- (D) Van der Waals

**Q26.**

1M [-0.33]

*[Textile Fibers]*

Cotton is a:

- (A) Protein
- (B) Cellulosic
- (C) Synthetic
- (D) Mineral

**Q27.**

1M [no -ve]

*[General Aptitude]*

Perimeter of square side 5:

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*Enter your answer here***Q28.**

1M [-0.33]

*[Apparel Engineering]*

Pattern is for:

- (A) Fabric design
- (B) Garment shape
- (C) Dyeing
- (D) Finishing

**Q29.**

1M [-0.33]

*[Textile Testing]*

Tensile strength measured by:

- (A) Microscope
- (B) Tensile tester
- (C) Air permeability
- (D) None

**Q30.**

1M [-0.33]

*[Textile Testing]*

Tensile strength measured by:

- (A) Microscope
- (B) Tensile tester
- (C) Air permeability
- (D) None

**Q31.**

2M [no -ve]

*[Textile Fibers]*

Cotton is a:

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*Enter your answer here***Q32.**

2M [-0.66]

*[Apparel Engineering]*

Pattern is for:

- (A) Fabric design
- (B) Garment shape
- (C) Dyeing
- (D) Finishing

**Q33.**

2M [no -ve]

*[Engineering Mathematics]*Integral of  $2x \, dx$ :

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*Enter your answer here***Q34.**

2M [-0.66]

*[Engineering Mathematics]*Integral of  $2x \, dx$ :

- (A)  $x^2 + C$
- (B)  $2x + C$
- (C)  $x + C$
- (D)  $x^{2/2}$

**Q35.**

2M [no -ve]

*[Textile Machinery]*

Ring frame produces:

- (A) Sliver
- (B) Roving
- (C) Yarn
- (D) Fabric

**Q36.**

2M [-0.66]

*[Textile Testing]*

Tensile strength measured by:

- (A) Microscope
- (B) Tensile tester
- (C) Air permeability
- (D) None

**Q37.**

2M [no -ve]

*[Yarn Manufacturing]*

Carding does:

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*Enter your answer here***Q38.**

2M [no -ve]

*[Yarn Manufacturing]*

Carding does:

- (A) Parallelize fibers
- (B) Spin fibers
- (C) Dye
- (D) Weave

**Q39.**

2M [-0.66]

*[Textile Machinery]*

Ring frame produces:

- (A) Sliver
- (B) Roving
- (C) Yarn
- (D) Fabric

**Q40.**

2M [-0.66]

*[Textile Machinery]*

Ring frame produces:

- (A) Sliver
- (B) Roving
- (C) Yarn
- (D) Fabric

**Q41.**

2M [-0.66]

*[Apparel Engineering]*

Pattern is for:

- (A) Fabric design
- (B) Garment shape
- (C) Dyeing
- (D) Finishing

**Q42.**

2M [-0.66]

*[Textile Testing]*

Tensile strength measured by:

- (A) Microscope
- (B) Tensile tester
- (C) Air permeability
- (D) None

**Q43.**

2M [-0.66]

*[Textile Physics]*Coulomb friction law:  $F = ?$ 

- (A)  $\mu N$
- (B)  $\mu^2 N$
- (C)  $N/\mu$
- (D)  $\mu^2 N$

**Q44.**

2M [-0.66]

*[Chemical Processing]*

Reactive dye bonds with:

- (A) Physical
- (B) Covalent
- (C) Hydrogen
- (D) Van der Waals

**Q45.**

2M [-0.66]

*[General Aptitude]*

Perimeter of square side 5:

- (A) 10
- (B) 20
- (C) 25
- (D) 5

**Q46.**

2M [no -ve]

*[Apparel Engineering]*

Pattern is for:

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*Enter your answer here***Q47.**

2M [-0.66]

*[Textile Physics]*Coulomb friction law:  $F = ?$ 

- (A)  $\mu R$
- (B)  $\mu \Delta R$
- (C)  $N/\mu$
- (D)  $\mu' \Delta$

**Q48.**

2M [no -ve]

*[Chemical Processing]*

Reactive dye bonds with:

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*Enter your answer here***Q49.**

2M [-0.66]

*[Engineering Mathematics]*Integral of  $2x \, dx$ :

- (A)  $x^2 + C$
- (B)  $2x + C$
- (C)  $x + C$
- (D)  $x^2/2$

**Q50.**

2M [no -ve]

*[Textile Physics]*Coulomb friction law:  $F = ?$ 

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*Enter your answer here***Q51.**

2M [-0.66]

*[Textile Machinery]*

Ring frame produces:

- (A) Sliver
- (B) Roving
- (C) Yarn
- (D) Fabric

**Q52.**

2M [no -ve]

*[Chemical Processing]*

Reactive dye bonds with:

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*Enter your answer here*

**Q53.**

2M [no -ve]

*[Engineering Mathematics]*Integral of  $2x \, dx$ :

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*Enter your answer here***Q54.**

2M [no -ve]

*[Textile Machinery]*

Ring frame produces:

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*Enter your answer here***Q55.**

2M [-0.66]

*[Fabric Manufacturing]*

Plain weave has:

- (A) 1 up 1 down
- (B) 2 up 2 down
- (C) 3 up 1 down
- (D) Random

**Q56.**

2M [-0.66]

*[Chemical Processing]*

Reactive dye bonds with:

- (A) Physical
- (B) Covalent
- (C) Hydrogen
- (D) Van der Waals

**Q57.**

2M [-0.66]

*[Textile Physics]*Coulomb friction law:  $F = ?$ 

- (A)  $\mu R$
- (B)  $\mu \omega$
- (C)  $N/\mu$
- (D)  $\mu' R$

**Q58.**

2M [no -ve]

*[Textile Physics]*Coulomb friction law:  $F = ?$ 

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*Enter your answer here***Q59.**

2M [-0.66]

*[Apparel Engineering]*

Pattern is for:

- (A) Fabric design
- (B) Garment shape
- (C) Dyeing
- (D) Finishing

**Q60.**

2M [-0.66]

*[General Aptitude]*

Perimeter of square side 5:

- (A) 10
- (B) 20
- (C) 25
- (D) 5

**Q61.**

2M [no -ve]

*[Textile Physics]*Coulomb friction law:  $F = ?$ 

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*Enter your answer here***Q62.**

2M [no -ve]

*[Textile Fibers]*

Cotton is a:

- (A) Protein
- (B) Cellulosic
- (C) Synthetic
- (D) Mineral

**Q63.**

2M [-0.66]

*[Yarn Manufacturing]*

Carding does:

- (A) Parallelize fibers
- (B) Spin fibers
- (C) Dye
- (D) Weave

**Q64.**

2M [no -ve]

*[Textile Fibers]*

Cotton is a:

- (A) Protein
- (B) Cellulosic
- (C) Synthetic
- (D) Mineral

**Q65.**

2M [no -ve]

*[Fabric Manufacturing]*

Plain weave has:

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*Enter your answer here*